

Sequences

Definition (convergence): A number $L \in \mathbb{R}$ is a limit of the sequence $\{a_n\}_{n=1}^{\infty}$ if for every $\epsilon > 0$ there exists an $N(\epsilon)$ such that for all $n > N(\epsilon)$, we have

$$|a_n - L| < \epsilon$$

A sequence that has a limit is said to converge. In such a case we write

$$a_n \rightarrow L \text{ and } n \rightarrow \infty$$

or equivalently

$$\lim_{n \rightarrow \infty} a_n = L$$

If a sequence does not converge to a finite value, we say it diverges.

Definition (monotonic): We say an infinite or finite sequence is strictly increasing if for all n

$$a_n < a_{n+1}$$

It is strictly decreasing if for all n

$$a_n > a_{n+1}$$

It is increasing if for all n

$$a_n \leq a_{n+1}$$

It is decreasing if for all n

$$a_n \geq a_{n+1}$$

Definition (bounded sequences): We say a sequence is bounded above if there is a number M , called an upper bound, such that for all $n \in \mathbb{N}$

$$a_n \leq M$$

It is bounded below if there is a number m , called a lower bound, such that for all $n \in \mathbb{N}$

$$a_n \geq m$$

Here are some simple statements that can be made from these definitions:

- $\{a_n\}_{n=1}^{\infty}$ being bound does not imply a_n has a limit
- $\{a_n\}_{n=1}^{\infty}$ has a limit does imply a_n is bounded
- $\{a_n\}_{n=1}^{\infty}$ being unbounded implies a_n has NO limit

Theorem. If an infinite sequence is monotonic increasing and bounded above, then it must converge to a finite limit.

Theorem (The Squeeze Theorem). Let $\{a_n\}_{n=1}^{\infty}$, $\{b_n\}_{n=1}^{\infty}$, and $\{c_n\}_{n=1}^{\infty}$ be sequences. Suppose that

$$a_n \leq b_n \leq c_n$$

and that

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n.$$

Then the sequence b_n converges and its limit is the same

as a_n and c_n .

Common limits for sequences:

$$\lim_{n \rightarrow \infty} r^n = \begin{cases} \infty & \text{if } |r| > 1 \\ 0 & \text{if } |r| < 1 \end{cases}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1, \text{ provided } a > 0$$

$$\lim_{n \rightarrow \infty} \frac{\ln(n)}{n^c} = 0, \quad \forall c > 0$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n = e^a, \text{ provided } a \in \mathbb{R}$$

$$\lim_{n \rightarrow \infty} \frac{n^c}{p^n} = 0, \quad \forall p > 1, \forall c > 0$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$$

$$\lim_{n \rightarrow \infty} \frac{c^n}{n!} = 0, \quad \forall c \in \mathbb{R}$$

Power Series

Definition (power series). A power series is a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

where $\{c_n\}_{n=0}^{\infty}$ is a sequence of real numbers

Definition (radius of convergence). Let $\sum_{n=0}^{\infty} c_n (x-a)^n$ be a power series and let

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|$$

The radius of convergence, R , of the series is equal to

$$(i) R = \frac{1}{\rho} \text{ when } 0 < \rho < \infty$$

$$(ii) R = 0 \text{ when } \rho = \infty$$

$$(iii) R = \infty \text{ when } \rho = 0$$

Theorem. Given a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$,

(i) the series converges for all x that satisfy $|x-a| < R$ when $R = \frac{1}{\rho}$

(ii) the series converges just for $x=a$ when $R=0$

(iii) the series converges for all x when $R=\infty$

Series

Definition (series). Given a sequence $\{a_n\}_{n=1}^{\infty}$, the corresponding series is the sequence $\{s_n\}_{n=1}^{\infty}$ of partial sums for which the n th term is

$$s_n = a_1 + a_2 + \dots + a_n = \sum_{k=1}^n a_k$$

We denote the full series by $\sum_{k=1}^{\infty} a_k$

Definition (telescoping sum). A finite sum $s_n = \sum_{k=1}^n a_k$ in which subsequent terms cancel each other, leaving only a finite number of terms is called a telescoping sum.

Theorem (The Harmonic Series). $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots$ diverges

Tests for convergence and divergence

• **Divergence test:** if $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ does not converge

• **Comparison test:** if $a_n \leq b_n$, then $\sum_{n=1}^{\infty} b_n$ converging implies $\sum_{n=1}^{\infty} a_n$ converges
the use, $\sum_{n=1}^{\infty} a_n$ diverging implies $\sum_{n=1}^{\infty} b_n$ diverges

• **Ratio test:** Let $K = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ such that $K < 1$ implies convergence, $K > 1$ implies divergence, and $K=1$ implies the test is inconclusive

• **Leibniz test:** The alternating series $\sum_{n=1}^{\infty} (-1)^n a_n$ converges if $a_n \geq 0 \forall n$, a_n is decreasing, and $\lim_{n \rightarrow \infty} a_n = 0$

• **Absolute convergence:** $\sum_{n=1}^{\infty} |a_n|$ converging implies $\sum_{n=1}^{\infty} a_n$ also converges (the series converges absolutely)